NLP-Based Sentiment Analysis System Using Naïve Bayes and Transformers

Problem Statement Document — Phase 1, Week 1

Project Area: Artificial Intelligence / Machine Learning / Natural Language Processing

Primary Focus: Machine-learning-based sentiment analysis of user-generated text reviews and comments

1. Background

Sentiment analysis is a major natural language processing task in which text is examined to determine the emotional tone or opinion expressed by its author, typically classified as positive, negative, or occasionally neutral. IEEE Access research on danmaku (bullet-screen) sentiment analysis describes sentiment analysis as a process of judging emotion expression through language, including extracting emotional information, processing and analyzing data, and classifying the sentiment of selected text. Such analysis is widely applied in public-opinion monitoring, business intelligence, e-commerce review analysis, and social media understanding.

Text-based sentiment analysis is an important research direction because a short text such as a review, comment, or post contains lexical, syntactic, and semantic information that can be extracted directly without requiring the system to access any external service or website. Existing IEEE research has investigated text features with traditional machine-learning models such as Naïve Bayes and SVM, dictionary-based methods, hybrid dictionary–machine-learning systems, and deep transformer-based models such as BERT and DeBERTa.

2. Problem Context

Traditional sentiment analysis mechanisms commonly include manually maintained sentiment dictionaries, lexicon-based scoring rules, and heuristic polarity detection. These mechanisms are useful, but they can struggle when text contains slang, emoticons, misspellings, sarcasm, or domain-specific vocabulary that is not present in the dictionary. Research has therefore explored machine-learning methods that learn patterns from labeled positive and negative text, and more recently transformer-based models that capture contextual meaning.

However, a machine-learning detector introduces its own challenges. A model can perform very well on the dataset used for training and testing while performing less effectively on text from a different source or domain. Recent IEEE Access research on aspect-based sentiment analysis specifically demonstrates the importance of cross-dataset validation: models trained on restaurant reviews showed clear performance degradation when evaluated on books, clothing, or hotel reviews, and even large language models struggled on datasets containing multiple aspects and mixed sentiment within a single sentence.

3. Core Problem Statement

Existing sentiment analysis approaches can achieve high classification performance under controlled experimental conditions, but their effectiveness may be limited by dataset dependency, domain shift, informal and colloquial text, implicit or mixed sentiment, class imbalance, feature selection, and reduced generalization to unseen or independently collected text. Therefore, there is a need for an AI-powered sentiment analysis system that systematically extracts meaningful textual characteristics, compares classical (Naïve Bayes) and transformer-based approaches, evaluates their performance using appropriate metrics, and investigates whether the resulting model generalizes beyond the dataset on which it was trained.

4. Research Problem

The research problem is not simply to classify a text as positive or negative. The project must investigate whether machine-learning models can provide reliable sentiment analysis while maintaining acceptable performance on previously unseen text and domains.

Challenge	Research concern
Dataset dependency	A model may learn vocabulary specific to one dataset (e.g., movie reviews) rather than general sentiment patterns.
Domain generalization	Performance can decrease when the model is evaluated on an independent dataset or a different domain (e.g., restaurant reviews evaluated on book reviews).
Informal and colloquial text	User-generated text contains slang, abbreviations, emoticons, and misspellings that standard vocabularies do not cover.
Implicit and mixed sentiment	A single sentence may express both positive and negative opinions toward different aspects (e.g., “great camera but terrible battery”).
Feature selection	Large text feature sets (lexical, TF-IDF, embeddings) may contain redundant or unstable features.
Class imbalance	Public review datasets are often dominated by positive examples, which can bias training.
False positives and false negatives	Incorrect classification can misrepresent user opinion or trigger wrong downstream business decisions.
Explainability	Users and researchers benefit from understanding which words or features

	influence a prediction.
Computational cost	Transformer models require significantly more training and inference resources than classical models such as Naïve Bayes.

5. Proposed Problem-Solving Direction

The proposed project will investigate a text-based AI/ML pipeline in which a submitted review or comment is validated, transformed into measurable lexical and semantic features, and passed to trained classification models. The system will produce a classification and a sentiment probability. Multiple algorithms can be compared, with a Naïve Bayes classifier using TF-IDF/dictionary-based features representing the classical approach and a fine-tuned transformer model (e.g., BERT) representing the contextual deep-learning approach. The hybrid dictionary–machine-learning strategy demonstrated in the danmaku sentiment study will inform feature design, and cross-dataset evaluation will follow the methodology of the aspect-based sentiment comparison study.

Proposed high-level workflow:

User Text → Text Validation → Preprocessing (cleaning, tokenization, stop-word removal) → Feature Extraction (TF-IDF / sentiment lexicon / embeddings) → Feature Selection/Preprocessing → ML Classifiers (Naïve Bayes + Transformer) → Sentiment Probability → Sentiment Classification → Result/Explanation

6. Project Objectives

1. Study existing sentiment analysis approaches through the selected research papers.
2. Identify important textual characteristics (lexical, syntactic, and semantic features) that can support sentiment classification, including sentiment dictionaries and catchwords for informal text.
3. Prepare and preprocess suitable positive and negative labeled text datasets (e.g., product or restaurant reviews, social media comments).
4. Implement and compare multiple classification approaches: a classical Naïve Bayes classifier and a transformer-based (e.g., BERT) classifier.
5. Evaluate models using accuracy, precision, recall, F1-score, specificity, ROC-AUC and confusion matrices where applicable.
6. Investigate model generalization using data that is independent from the training set, including cross-domain evaluation.
7. Provide a usable interface through which a user can submit a text review or comment for analysis.
8. Provide an interpretable result containing classification, probability/risk information and relevant feature (word) information where possible.

7. Scope of the Problem

In scope: text validation, text preprocessing, feature extraction (TF-IDF, sentiment lexicon features, contextual embeddings), feature selection, supervised machine learning (Naïve Bayes and transformer-based models), model comparison, evaluation, cross-dataset validation, prediction probability, sentiment classification, and a web-based demonstration interface.

Initially out of scope: real-time streaming analysis at large scale, full aspect-level sentiment extraction, multi-language support beyond the selected dataset language, guaranteed detection of sarcasm and implicit sentiment, and integration with live social media platforms. These may be considered future work.

Reference note: The problem formulation is intentionally broader than a single algorithm. Students should use the literature to justify their eventual choice of dataset, features, models and evaluation strategy.